

Dynamic Platoon Formation of Multi-Type Autonomous Vehicles for Sustainable Urban Mobility

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Abstract

We study an algorithmic abstraction for future cooperative autonomous mobility in which larger active vehicles can tow smaller passive vehicles along shared road segments. The core problem is spatio-temporal: given fixed vehicle trajectories, entry times, speeds, and active-vehicle towing capacities, determine which passive vehicles should be matched to which active vehicles, and over which route segments, to maximize total covered distance. Unlike standard platooning or one-to-one assignment, the problem allows multi-segment service, where a passive vehicle may be towed by different active vehicles over disjoint portions of its route, while active-vehicle capacity must be respected at every position along the corridor. We formulate this dynamic platoon formation problem and propose two approximation methods: a greedy maximum-weight matching baseline and an Iterative Linear Assignment method that expands active vehicles into virtual capacity slots and repeatedly solves linear sum assignment subproblems. Experiments on synthetic corridor scenarios with up to 400 passive and 80 active vehicles show that both methods achieve substantial towed-distance coverage (25–52% depending on capacity), with the assignment-based method consistently matching or exceeding greedy solution quality while reducing empirical runtime by 2.8–11.5×. We position the model as a corridor-level spatial matching primitive; mechanical coupling, safety, and full road-network routing are left as future extensions.

CCS Concepts

• **TODO: replace with CCS-generated concept → e.g. Theory of computation.**

Keywords

autonomous vehicles, vehicle platooning, spatial-temporal matching, linear assignment, sustainable mobility

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1 Introduction

Single-occupancy travel remains a major barrier to sustainable urban mobility. U.S. National Household Travel Survey data show that drive-alone commuting dominates trip demand [?], while the transportation sector contributes about 29% of U.S. greenhouse gas emissions [?]. This inefficiency is pronounced when many vehicles traverse overlapping routes but still consume propulsion energy independently. With rapid advances in autonomous driving [?], future transportation can realize physically coordinated vehicle platoons that go beyond independent driving.

In a near-future setting where vehicles are fully autonomous, new forms of physical coordination between vehicles become feasible: a large autonomous vehicle could mechanically couple with smaller vehicles on a shared road segment, towing them and offsetting a substantial share of their independent propulsion energy [?]. Realizing such coordinated interaction at scale requires *systematic decisions* about which vehicles should couple, when, and for how long along their routes. We therefore treat mechanical coupling, safety, and regulatory design as system assumptions outside the scope of this short paper, and focus on the spatio-temporal matching layer that would be required if such coupling were available.

To address this challenge we introduce a new problem that we call *dynamic platoon formation*. We consider two classes of autonomous vehicles: larger *Active Vehicles (AVs)* capable of physically towing smaller vehicles, and smaller *Passive Vehicles (PVs)* that can be attached to AVs during shared road segments. Unlike drafting-based platooning, which yields modest aerodynamic savings while vehicles remain physically separated, our model assumes mechanical coupling: while a PV is attached to an AV we model the PV as not self-propelled over that segment, with the segment's towing load attributed to the AV. Given a set of AVs and PVs with known, fixed routes (AVs do not deviate from their pre-planned paths), entry times, and speeds, the goal is to **maximize the total PV distance covered by towing**. We use towed distance as a proxy for PV-side propulsion-energy reduction and reserve physics-level load modeling for future work.

Figure 1 illustrates this on a shared corridor with two AVs and three PVs. Two features are central to our formulation: *point-wise capacity*—between $x=20$ and $x=60$, A_1 simultaneously tows T_1 and T_2 , reaching its capacity of 2—and *multi-segment matching*, where

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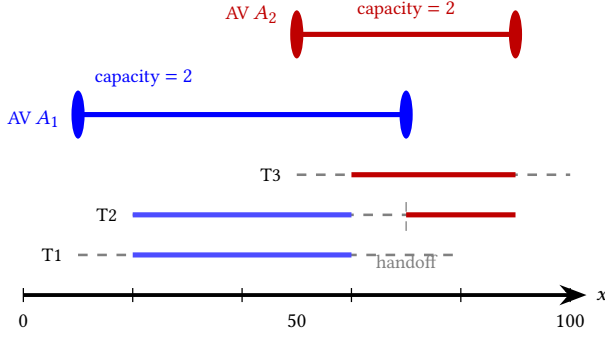


Figure 1: Conceptual overview of towing-based cooperation on a 1D corridor (positions 0–100 m). Dashed gray segments denote self-driving portions; solid colored segments denote towed travel, on which the PV is modeled as not self-propelled. AVs A_1 and A_2 each have towing capacity 2. T_2 is towed by A_1 over $[20, 60]$, self-drives over $[60, 70]$, and is handed off to A_2 over $[70, 90]$ —an instance of *multi-segment matching*.

T_2 is served by A_1 and A_2 on disjoint portions of its route with a self-driving gap in between.

Prior work spans three threads. (i) Platooning research focuses on aerodynamic coordination among physically separated vehicles via cooperative adaptive cruise control [? ? ? ?]. (ii) Ridesharing and dial-a-ride methods match passengers or trips to vehicles under spatial and temporal constraints, but the decision variable is passenger placement rather than vehicle-to-vehicle coupling along route segments [? ? ? ?]. (iii) Combinatorial assignment provides algorithmic tools for one-to-one matching, including the Hungarian method and its extensions [? ?]. None of these lines captures the combination of features required for dynamic towing coordination: sustained physical coupling, multi-segment coverage of a single PV by different AVs, and *point-wise* AV capacity constraints—the requirement that the number of PVs an AV tows simultaneously is enforced at every position along its route and can change as PVs attach and detach.

We formulate the problem as: given AVs and PVs with fixed routes, entry times and speeds, decide which PVs attach to which AVs over which road segments to maximize the total towed PV distance, subject to spatial overlap requirements, point-wise capacity, and temporal synchronization at coupling points. This yields a state-dependent combinatorial matching problem that differs from a static Generalized Assignment Problem (GAP) [?], because assigning one towing segment can change later temporal feasibility for the same PV. Exact global solvers are therefore impractical at fleet scale, and we propose two approximations. The first, a *Greedy Maximum-Weight Matching*, repeatedly commits to the highest-saving feasible AV–PV segment until no feasible pair remains—fast but myopic. The second, an *Iterative Linear Assignment* (ILA) algorithm, takes a globally coordinated view: in each round it expands every AV into virtual capacity slots, builds a bipartite graph of feasible AV–slot–PV pairings, and solves the resulting *Linear Sum Assignment Problem* (LSAP) optimally via the Jonker–Volgenant method [?]. PV states

Table 1: Summary of notation. Positions and lengths are in meters, times in seconds, speeds in m/s. Capacities are dimensionless integer counts.

Symbol	Description (unit)
L	Corridor length (m)
N, M	Number of AVs and PVs (dimensionless)
e_i^a, x_i^a	Entry/exit position of AV a_i (m)
e_j^p, x_j^p	Entry/exit position of PV p_j (m)
C_i	Towing capacity of AV a_i ($\in \mathbb{Z}_{\geq 0}$, count)
t_i^a, v_i^a	Entry time (s) and constant speed (m/s) of a_i
t_j^p, v_j^p	Entry time (s) and self-driving speed (m/s) of p_j
cp_{ij}, dp_{ij}	Earliest possible coupling / latest decoupling positions of (a_i, p_j) (m)
d_{ij}	Shared overlap distance of (a_i, p_j) (m)
$L_{\min}$	Minimum towing-segment length (m)
τ	Temporal coupling tolerance (s)

are then updated to reflect newly covered segments and the round repeats until no further assignment is possible. For tractability we evaluate both algorithms on a one-dimensional corridor abstraction; we discuss extensions to 2D road networks in Section 5.

Contributions. (1) We formalize *dynamic platoon formation* as a spatio-temporal matching problem with multi-segment service and point-wise AV capacity, instantiated on a 1D corridor. (2) We develop and compare two approximation algorithms—a greedy maximum-weight baseline and an iterative LSAP-based method with virtual-slot capacity encoding. (3) On synthetic corridor scenarios with up to 80 AVs and 400 PVs, ILA matches or exceeds greedy solution quality (25–52% towed-distance coverage depending on capacity) while running 2.8–11.5× faster.

2 Problem Formulation

We formalize *dynamic platoon formation* on a one-dimensional corridor of length L . Restricting attention to a single corridor isolates the spatio-temporal matching structure—segment overlap, point-wise capacity, and temporal synchronization—from routing decisions; extending the formulation to 2D road networks is discussed in Section 5. Throughout, positions are in meters (m), times in seconds (s), and speeds in m/s; towing capacities are nonnegative integers.

2.1 Vehicles and Notation

A problem instance consists of two finite sets: N *Active Vehicles* (AVs) $\mathcal{A} = \{a_1, \dots, a_N\}$ and M *Passive Vehicles* (PVs) $\mathcal{P} = \{p_1, \dots, p_M\}$. Each AV a_i is described by the tuple $(e_i^a, x_i^a, C_i, t_i^a, v_i^a)$ with entry/exit positions $e_i^a, x_i^a \in [0, L]$ (where $e_i^a < x_i^a$), a scalar towing capacity $C_i \in \mathbb{Z}_{\geq 0}$, an entry time t_i^a and a constant speed $v_i^a > 0$. Each PV p_j is described by $(e_j^p, x_j^p, t_j^p, v_j^p)$, defined analogously, but has no towing capacity of its own. Table 1 summarizes the symbols. We emphasize that C_i is a *scalar* per-AV bound, not a vector: the capacity ranges varied in Section 4 are values swept *across* instances, not per-route vectors for a single AV.

2.2 Feasibility of a Towed Segment

A candidate towing segment is a tuple (a_i, p_j, s, e) with $0 \leq s < e \leq L$, meaning that p_j is towed by a_i over the spatial interval $[s, e]$.

The maximum geometrically realizable overlap of routes (a_i, p_j) is the closed interval $[cp_{ij}, dp_{ij}]$, where $cp_{ij} = \max(e_i^a, e_j^p)$ is the earliest possible coupling position and $dp_{ij} = \min(x_i^a, x_j^p)$ the latest possible decoupling; selected s, e may satisfy $cp_{ij} \leq s < e \leq dp_{ij}$. The overlap length is

$$d_{ij} = \max(0, dp_{ij} - cp_{ij}). \quad (1)$$

A segment (a_i, p_j, s, e) is *feasible* if (i) $[s, e] \subseteq [cp_{ij}, dp_{ij}]$ (spatial inclusion), (ii) $e - s \geq L_{\min}$ (minimum length, justifying coupling overhead), and (iii) the two vehicles reach the segment's starting point within tolerance τ :

$$|T_i^a(s) - T_j^p(s)| \leq \tau, \quad (2)$$

where $T_i^a(x) = t_i^a + (x - e_i^a)/v_i^a$ is AV a_i 's arrival time at position x . The PV arrival time $T_j^p(s)$ is *state-dependent*: for an initially uncovered PV it is the self-driving time $t_j^p + (s - e_j^p)/v_j^p$, but after p_j has been towed on an earlier segment, our algorithms (Section 3) update its route-state time before evaluating later segments. We use the towed distance $e - s$ directly as the saving proxy on a feasible segment, reserving energy-level modeling that accounts for AV-side towing load for future work.

2.3 Multi-Segment Matching and Optimization

Let $\mathcal{X}$ denote a set of feasible segments selected by the algorithm. We allow a single PV to be served by multiple AVs along disjoint portions of its route—*multi-segment matching*—but the same PV cannot be towed by two AVs at the same position. The optimization problem is

$$\max_{\mathcal{X}} \sum_{(a_i, p_j, s, e) \in \mathcal{X}} (e - s) \quad (3)$$

$$\text{s.t. } |\{(a_i, p_j, s, e) \in \mathcal{X} : x \in [s, e)\}| \leq C_i \quad (4)$$

$$\forall a_i \in \mathcal{A}, \forall x \in [e_i^a, x_i^a] \quad (4)$$

$$\text{Segments assigned to each } p_j \text{ are pairwise disjoint} \quad (5)$$

$$\text{Each } (a_i, p_j, s, e) \in \mathcal{X} \text{ is feasible (Sec. 2.2).} \quad (6)$$

The objective (3) is the total towed distance across all selected segments. Constraint (4) is the *point-wise capacity* requirement: at every position x along an AV's route, the number of PVs currently attached to it is bounded by C_i . Constraint (5) forbids assigning two AVs to the same PV over overlapping positions.

Problem structure. The problem has a combinatorial assignment structure with *state-dependent* feasibility: selecting one towing segment changes a PV's residual route timing and can affect the temporal feasibility of later segments, distinguishing it from a static GAP-style assignment [?]. We therefore do not pursue an exact global solver in this short paper and instead study scalable approximation algorithms in Section 3.

Critical-points reduction. Although constraint (4) ranges over a continuum of positions x , the count of attached PVs is a step function that changes only at the endpoints of selected segments. It therefore suffices to enforce capacity at the finite set of *critical points* $C_i = \{e_i^a, x_i^a\} \cup \{s, e : (a_i, p_j, s, e) \in \mathcal{X}\}$ for each AV a_i . Both algorithms in Section 3 exploit this reduction so that each feasibility check costs $O(|C_i|)$ rather than continuous verification.

2.4 Modeling Assumptions

For tractability and to keep the contribution focused on the matching layer, we adopt the following assumptions; relaxing each is left to future work.

- **Fixed routes and speeds.** AV and PV routes, entry times, and speeds are known and deterministic, and AVs do not re-route to attract PVs.
- **Coupled motion.** While a PV is attached to an AV, the PV follows the AV's speed; decoupling occurs at the end of the assigned segment or when either vehicle exits the corridor.
- **Negligible coupling overhead.** Coupling and decoupling are modeled as instantaneous and cost-free at the algorithmic layer; we partly absorb the physical overhead through the minimum-segment constraint $L_{\min}$.
- **Invariant demand.** The total set of vehicles is unchanged by matching—only *which* vehicle tows *which* changes—so we do not model induced congestion within this short paper.
- **Out-of-scope physical layer.** As noted in Section 1, mechanical coupling, safety, and regulatory design are treated as system-level assumptions outside the scope of this work.

3 Proposed Algorithms

Both algorithms maintain a PV *route-state* $(\ell_j, \hat{t}_j)$ where ℓ_j is the start of the unprocessed suffix of p_j 's route (initially e_j^p) and $\hat{t}_j$ its arrival time at ℓ_j (initially t_j^p). When p_j is committed to a segment $[s, e]$ on AV a_i , the state advances by $\ell_j \leftarrow e$ and $\hat{t}_j \leftarrow \hat{t}_j + (s - \ell_j^{\text{old}})/v_j^p + (e - s)/v_i^a$, reflecting self-driving to s followed by towing at the AV's speed to e . The state-dependent arrival time T_j^p in Eq. (2) reads from this $\hat{t}_j$.

3.1 Greedy Maximum-Weight Matching

At each iteration the greedy algorithm enumerates all feasible candidate segments (a_i, p_j, s, e) under the current PV states, selects the segment with the largest towed distance $e - s$, commits it, and updates the PV route-state; point-wise capacity is checked at the finite critical points of Section 2.3. The procedure terminates when no feasible segment remains. Each iteration generates and sorts $O(NM)$ candidates in $O(NM \log NM)$, and with up to $O(NM)$ commitments the total worst-case complexity is $O((NM)^2 \log NM)$. This baseline is fast but myopic: committing one high-saving segment can block a co-optimal alternative.

3.2 Iterative Linear Assignment (ILA)

ILA replaces the myopic commit with a sequence of *global* matching rounds. In each round, every AV a_i is replicated into C_i identical *virtual slots*, encoding per-AV multiplicity in the LSAP while leaving the position-dependent capacity check to the commit step. With $K = \sum_i C_i$ slot nodes on the left and M PV nodes on the right, we form a cost matrix $W \in \mathbb{R}^{K \times M}$ whose entry for slot k of AV a_i and PV p_j is $-d_{ij}$ for a feasible pair and $+\infty$ otherwise, where $d_{ij} = \max(0, dp_{ij} - \max(e_i^a, \ell_j))$ is the residual overlap that uses ℓ_j in place of e_j^p . Minimum-cost matching on W therefore maximizes total towed distance for the current round.

Algorithm 1 Iterative Linear Assignment (ILA)**Require:** AVs $\mathcal{A}$, PVs $\mathcal{P}$, parameters $L_{\min}, \tau$ **Ensure:** Selected segments $\mathcal{X}$

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1: Initialize  $(\ell_j, \hat{t}_j) \leftarrow (e_j^p, t_j^p)$  for all  $p_j$ ;  $\mathcal{X} \leftarrow \emptyset$ 
2: repeat
3:   Expand  $\mathcal{A}$  into  $K = \sum_i C_i$  virtual slots; build cost matrix  $W$ 
4:    $\mathcal{M} \leftarrow \text{JONKERVOLGENANT}(W) \triangleright$  per-round LSAP optimum
5:   for each pair  $(slot: a_i, p_j) \in \mathcal{M}$  with  $W_{slot,j} < +\infty$  do
6:     Let  $s \leftarrow \max(e_i^a, \ell_j)$ ,  $e \leftarrow dp_{ij}$ 
7:     if  $(a_i, p_j, s, e)$  feasible given prior commits this round
8:       then
9:          $\mathcal{X} \leftarrow \mathcal{X} \cup \{(a_i, p_j, s, e)\}$ ; update  $\ell_j, \hat{t}_j$ 
10:      end if
11:   end for
12: until no new pair was added in the last round
13: return  $\mathcal{X}$ 

```

Each round (Alg. 1) solves this Linear Sum Assignment Problem (LSAP) optimally with the Jonker–Volgenant algorithm [? ?]. Returned pairs are then sequentially confirmed against point-wise capacity at the critical points; pairs that fail (because multiple

slots of the same AV matched a non-disjoint set of segments in one round) are discarded and remain eligible in later rounds. Confirmed pairs advance the corresponding PV route-states, so the next round operates on the residual problem. Rounds repeat until no feasible match remains, yielding multi-segment service whenever the same PV is matched to different AVs across rounds. Jonker–Volgenant guarantees per-round optimality on the constructed bipartite sub-problem; global optimality across rounds is not guaranteed, but our experiments (Section 4) show ILA matches or exceeds greedy on every tested scenario. Each round runs in $O(KM + \max(K, M)^3)$, and at most $O(NM)$ rounds are required since each commit strictly advances some ℓ_j , giving overall complexity $O(NM \cdot \max(K, M)^3)$.

4 Experimental Evaluation

4.1 Setup

4.2 Results

5 Conclusion and Future Work

References